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Introduction

In the evolution of modern derivatives pricing, capturing the dynamics of volatility and extreme market events is essential for realistically valuing derivatives and managing risk effectively. In our previous project, we had established a foundation with discrete-time methods for vanilla option pricing, and now, our next step is to move beyond the constant volatility assumption of the Black-Scholes framework and investigate models that are a better reflection of how the market actually behaves.

Here, we focus on two advanced approaches: the Heston Stochastic Volatility Model and the Merton Jump-Diffusion Model. They both extend the limitations of Black-Scholes by incorporating additional sources of uncertainty. The Heston model allows volatility itself to evolve stochastically, while the Merton model introduces sudden jumps in the underlying asset price, which provides a more accurate description of markets' subjection to shocks, discontinuities, or fat-tailed distributions as we observe in reality.

Our task is to apply these models to pricing European and American options and analyzing the influence of stochastic volatility and jumps. Particular emphasis will be placed on how these features alter the Greeks (Delta and Gamma only).

Through this exercise, we aim to:

- Demonstrate the implementation of stochastic volatility and jump-diffusion frameworks for option pricing.
- Assess how volatility clustering and price jumps affect option valuation.
- Explore how these models affect risk sensitivities and hedging decisions.
- Establish a comparative understanding of when and why these models provide advantages over simpler approaches.

By expanding our toolkit from deterministic to stochastic and discontinuous volatility structures, we strengthen our desk's ability to handle the challenges of real-world option markets.

Section 1. Stochastic Volatility and Jump Diffusion Models (European Options)

1.1 Heston Model (Stochastic Volatility)

The Heston Model (1993) is a stochastic volatility model that extends the classical Black–Scholes framework by allowing the variance of the asset price to be random and mean-reverting, rather than constant. This makes it more realistic for capturing volatility clustering and the implied volatility smile observed in financial markets.

Model Dynamics

The asset price S_t and variance v_t follow the coupled SDEs under the risk-neutral measure:

$$dS_t = S_t\left(rdt + \sqrt{v_t}dW_t^S\right)$$

$$dvt = \kappa(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^v$$

with correlation:

$$dW_t^S dW_t^v = \rho dt$$

Parameters

- S_t**:** Asset price at time t
- v_t: Instantaneous variance of the asset
- **r:** Risk-free interest rate
- **κ:** Speed of mean reversion of the variance
- **θ:** Long-run mean of variance
- **σ:** Volatility of volatility (vol-of-vol)
- **ρ:** Correlation between the asset return shocks and variance shocks

- W_t^S, W_t^v Two Brownian motions (correlated)

1.1.1 Pricing ATM European Call and Put

Using the Heston Model and Monte-Carlo simulation, we priced an ATM European call and an ATM European put, using a correlation value of -0.30 & 0.70, and the following parameters - S_0 = 80; r = 5.5%; sigma = 35%; time to maturity = 3 months; v_0 = 3.2%; k_v = 1. 85; θ_v = 0. 045. The code begins by setting a random seed to keep the results reproducible, and then establishes the model's key ingredients: the starting variance, the long-run variance level, the speed of mean reversion, and the volatility of volatility.

With these in place, the simulation generates two streams of random shocks that are correlated, one to drive stock price movements and the other to drive variance changes. The stock evolves forward in time using a log-Euler step that reflects both the drift from the risk-free rate and the random effect of volatility, while the variance itself follows a mean-reverting process and is kept strictly non-negative through truncation.

Once many paths are simulated to maturity, the option payoff is applied at the end of each path — max (S_T − K, 0) for the call, or max (K − S_T, 0) for the put and then each payoff is discounted back to the present. Averaging across all simulated paths gives the option price, while the spread of results provides a measure of sampling error. The routine is flexible enough to handle both calls and puts by simply switching the payoff function, and it also saves the random shocks so that the same paths can be reused when estimating Greeks, which keeps numerical noise under control.

For the Heston model with correlation $\rho=-0.30$, the at-the-money European call option was priced at $3.41 with a standard error of about 0.014, while the corresponding put came to $2.32 with a standard error of 0.013. When the correlation was made more negative at $\rho=-0.70$, the call price increased slightly to 3.43 and the put to 2.35, both with similar error margins.

These results are consistent with intuition: stronger negative correlation between asset returns and volatility (the "leverage effect") leads to slightly higher option values, since volatility tends to rise when the asset falls, raising the likelihood of option payoffs. The Monte Carlo standard errors are small in each case, confirming that the simulations converged well.

1.1.2 Delta and Gamma under Stochastic Volatility

The Greeks were estimated using a simple bump-and-revalue approach with common random numbers. The stock price was nudged up and down by 1%, then we re-priced the option each time, and then computed the Delta as the slope of the price change, while Gamma came from the curvature across the bumped values.

For $\rho = -0.30$, the call showed a Delta of 0.62 and the put –0.38, with both options sharing a Gamma of about 0.058. When correlation was more negative at $\rho = -0.70$, the call Delta rose to 0.66 and the put Delta moved up to –0.34, while Gamma dropped slightly to 0.051. These results suggest that stronger negative correlation makes the call more sensitive to how the underlying price moves.

1.2 Merton Model (Jump-Diffusion)

The Merton Jump-Diffusion Model (1976) extends the classical Black–Scholes model by incorporating sudden and discontinuous movements (jumps) in asset prices, in addition to the usual continuous diffusion process. This makes it more realistic for capturing market events such as crashes, large announcements, or unexpected events.

Model Dynamics

The asset price S_t under the Merton jump-diffusion model evolves as:

$$dS_t = S_t(\mu dt + \sigma dW_t) + S_t dJ_t$$

Where:

$$J_t = \sum_{i=1}^{N_t} . (Y_i - 1)$$

and N_t is a Poisson process with intensity λ.

Equivalently, the log-price dynamics are:

$$dlnS_t = \left(\mu - \frac{1}{2}\sigma^2 - \lambda k\right) dt + \sigma dW_t + \sum_{i=1}^{N_t} lnY_i$$

Parameters

- S_t: Asset price at time t
- μ: Drift term (expected rate of return)
- σ: Volatility of the continuous part (diffusion)
- W_t: Standard Brownian motion
- N_t: Poisson process with rate λ (average number of jumps per unit time)
- Y_i: Random jump size
- λ: Jump intensity (frequency of jumps)
- k=E[Y−1]: Mean percentage jump size adjustment

1.2.1 Pricing an ATM European Call and Put

Using the parameters S_0=80; K=80; r=5.5%, and T = 3 months, we priced options using the Merton model. The simulation starts by fixing the diffusion volatility (σ=35%) and the jump parameters: the mean and volatility of log jump sizes, along with the jump intensity (λ), which controls how frequently jumps are expected.

The path generator works by stepping forward in small increments. At each step, two things happen:

- a diffusion component makes the stock price evolve smoothly, using exponential update with Gaussian shocks,

- and a jump component is added, based on Poisson draws for the number of jumps during the step, and each jump size is drawn from a lognormal distribution.

These two components combine so that the stock evolves with both regular volatility and occasional, discrete shocks up or down. Repeating this over many paths produces a distribution of terminal stock prices that reflect both normal movements and also rare jumps.

To obtain option prices, the routine applies the standard payoff at maturity — max $(S_T - K, 0)$ for calls and max $(K - S_T, 0)$ for puts and discounts the average back at the risk-free rate. By running the experiment for two different jump intensities (λ=0.75 and λ=0.25), we can see how more frequent jumps influence the value of at-the-money options.

For the Merton jump-diffusion model with a higher jump intensity (λ=0.75), the at-the-money European call was priced at \$8.34 and the corresponding put at \$7.26. When the jump intensity was reduced to λ=0.25, the call price fell to \$6.89 and the put to \$5.79.

The difference between the two cases shows how impactful jumps can be. With more frequent jumps, the distribution of possible outcomes is wider, which increases the probability of extreme moves in either direction. This higher uncertainty leads to higher option values for both calls and puts. But with fewer jumps, the range payoff is not so wide and so the option prices are also lower.

1.2.3 Delta and Gamma under Jump-Diffusion

We used the same bump-and-revalue idea as before. We draw one set of Gaussian shocks (for diffusion) and Poisson jump counts (for jumps), then price the option three times on the same path set - at $S_0 - h$, S_0, and $S_0 + h$. Delta is the central slope of price vs. spot, and Gamma is the curvature from those three prices. This mirrors the Heston workflow, but Merton's jump dynamics was added.

Results (ATM, $K=S_0$)

- **λ = 0.75:** Call $\Delta \approx 0.649$, $\Gamma \approx 0.0225$; Put $\Delta \approx -0.352$, $\Gamma \approx 0.0222$.

- **λ = 0.25:** Call Δ≈0.598, Γ≈0.0266; Put Δ≈−0.402, Γ≈0.0258.

Interpretation

- **Deltas:** Calls are positive and puts negative, as expected. With higher jump intensity (λ=0.75), call Delta moves closer to 0.65 (smaller put), reflecting the fatter-tailed upside/downside mix.
- **Gammas:** Call and put Gammas are (near-)equal, consistent with vanilla parity. Higher λ slightly reduces Gamma (≈0.022 vs. 0.026), which is intuitive: jumps add tail risk that dilutes the local curvature captured by Gamma.

1.3 Model Validation

1.3.1 Put-Call Parity in Heston and Merton Models

We checked for put-call parity for the options we priced using the Heston and Merton model. Table 1.1 below presents the results.

Table 1.1 Put-Call Parity for Models

Model	Parameter	Call Price	Put Price	C − P	RHS (S_0 − K·e^(−rT))	Difference
Heston	ρ = −0.30	3.4354	2.3255	1.1098	1.0925	+0.0174
Heston	ρ = −0.70	3.4638	2.3232	1.1407	1.0925	+0.0482
Merton	λ = 0.75	8.2859	7.2433	1.0427	1.0925	−0.0498
Merton	λ = 0.25	6.8341	5.7073	1.1267	1.0925	+0.0343

From the table above, we can see that put-call parity holds in all four models, but it's tightest for Heston (ρ = −0.30). The larger gaps (up to ~0.05) in Heston (ρ = −0.70) and Merton cases suggest ordinary Monte-Carlo sampling noise (and are typical when jumps increase variance).

1.3.2 Strike Price Sensitivity and Moneyness Analysis

We did both the Heston Model and Merton Model for 7 different strikes: 3 OTM calls; 1 ATM call; and 3 ITM calls, using moneyness values: 0.85, 0.90, 0.95, 1, 1.05, 1.10, and 1.15.

The results for this experiment using the Heston model are presented in table 1.2 below

Table 1.2 Strike Price Sensitivity and Moneyness for Heston Model

ρ	Moneyness (S/K)	Strike K	Call Price	SE(Call)	Put Price	SE(Put)
-0.70	0.85	94.12	0.0315	0.0012	12.8467	0.0225
-0.70	0.90	88.89	0.2790	0.0035	7.9669	0.0215
-0.70	0.95	84.21	1.3441	0.0078	4.4056	0.0184
-0.70	1.00	80.00	3.4431	0.0123	2.3531	0.0143
-0.70	1.05	76.19	6.1026	0.0159	1.2488	0.0106
-0.70	1.10	72.73	8.9484	0.0184	0.6859	0.0078
-0.70	1.15	69.57	11.7257	0.0200	0.3616	0.0056
-0.30	0.85	94.12	0.1422	0.0033	12.9626	0.0222
-0.30	0.90	88.89	0.5207	0.0062	8.2104	0.0204

-0.30	0.95	84.21	1.5176	0.0104	4.5680	0.0172
-0.30	1.00	80.00	3.4289	0.0145	2.3177	0.0131
-0.30	1.05	76.19	5.9677	0.0178	1.1280	0.0093
-0.30	1.10	72.73	8.7929	0.0200	0.5498	0.0065
-0.30	1.15	69.57	11.6553	0.0214	0.2677	0.0044

We noticed that the results follow the expected pattern across strikes:

- **Call prices rise** as the strike decreases (deeper in-the-money), moving from near-zero at high strikes (OTM) to over 11 when the option is deep ITM.
- **Put prices fall** as the strike decreases, being largest when the strike is high (deep ITM puts worth ~13) and going near-zero as the strike drops below the spot.
- **Correlation effect:** Comparing ρ=−0.70 and ρ=−0.30, calls are slightly more expensive and puts are slightly cheaper when the negative correlation is stronger, which is consistent with the leverage effect (greater negative correlation amplifies volatility when prices fall, favouring calls slightly more) (Odhiambo, Murori and Aringo, 2025).
- **Standard errors** are small throughout, confirming stable Monte Carlo estimates.

Overall, the pricing surfaces behave consistently with no-arbitrage intuition and reflect the sensitivity of option values to both strike and volatility correlation.

The same was done using the Merton model for pricing and the results are summarized in table 1.3 below

Table 1.3 Strike Sensitivity and Moneyness for Merton Model

λ	**Moneyness (S/K)**	**Strike K**	**Call Price**	**Put Price**	**Parity Diff**

0.25	0.85	94.12	1.9994	14.9404	−0.1086
0.25	0.90	88.89	3.2728	10.9158	+0.0320
0.25	0.95	84.21	4.9201	7.9538	+0.0269
0.25	1.00	80.00	6.8450	5.6889	+0.0636
0.25	1.05	76.19	8.9541	4.1196	−0.0155
0.25	1.10	72.73	11.2269	2.9841	−0.0231
0.25	1.15	69.57	13.5616	2.2223	−0.0455
0.75	0.85	94.12	2.7766	15.6420	−0.0330
0.75	0.90	88.89	4.3729	12.0388	+0.0092
0.75	0.95	84.21	6.2381	9.1712	+0.1275
0.75	1.00	80.00	8.3257	7.1978	+0.0355
0.75	1.05	76.19	10.5525	5.6848	+0.0178
0.75	1.10	72.73	12.8892	4.5719	+0.0515
0.75	1.15	69.57	15.1320	3.6935	+0.0537

For the Merton model, we also noticed a few things:

- **Strike dependence:** As in the Heston case, calls increase and puts decrease as the strike moves deeper into the money. Deep ITM calls reach values above 15, while deep ITM puts remain in the 2–4 range.
- **Jump intensity effect:** Comparing λ = 0.25 vs λ = 0.75, option prices are systematically

higher when jump intensity is larger. This reflects the broader distribution of outcomes when jumps are more frequent; more tail risk means both calls and puts are more valuable.

- **Put–call parity:** Differences are small overall, typically within ±0.05–0.12, consistent with Monte Carlo error. Slightly larger deviations appear around $\lambda = 0.75$ and strikes near 0.95 moneyness, which is expected since jumps introduce more variance and increase noise in parity checks.

Overall, the results confirm the expected shape of option prices across strikes, while also highlighting how higher jump intensity in the Merton model inflates both call and put valuations and makes parity checks a bit noisier.

Section 2: American Options Under Stochastic Volatility

2.1 American Options under Stochastic Volatility

We went on to price an American option using Heston model with correlation = -0.30 with the help of Longstaff-Schwartz method.

Longstaff–Schwartz (LSM) prices early-exercise options by estimating the continuation value with a regression:

$$Et[\,disc \times future\ cashflow \mid S_t\,] \approx \beta \top \phi(S_t),$$

where $\phi(S_t)$ are simple basis functions (e.g., 1, S_t, S_t^2). If the immediate exercise payoff exceeds this estimated continuation value, the path exercises at t; otherwise it continues.

Why did we use Longstaff–Schwartz? LSM method is good for pricing American options under the Heston stochastic volatility model because American options do not have a closed form under Heston (Samimi & Mehrdoust, 2018). American options require evaluating whether to exercise or continue at multiple points in time, which can become a high-dimensional problem due to the process depending on price and also volatility. Traditional PDE or lattice approaches suffer from the curse of dimensionality and can be computationally expensive, while Monte Carlo simulation naturally accommodates multiple stochastic factors. We can simply do regression on simulated paths to estimate continuation values, combining Monte Carlo and LSM.

We start by simulating a large set of Heston price paths with $\rho=-0.30$ over the 3-month horizon. At maturity, each path's payoff is just max (ST − K, 0). Then we walk backward in time: at each date, we look only at in-the-money paths, regress their discounted future cashflows on $[1, S_t, S_t^2]$ to estimate what they're worth if held, and compare that estimate to the instant exercise value max $(S_t - K, 0)$. If exercising dominates, we take the cashflow there and stop that path; if not, we keep its future cashflow. After sweeping back to the start, we discount each exercised cashflow to time zero and average across paths to get the American price. For context, we also price the European call on the same model by simply taking the discounted payoff at maturity; no early-exercise step so we can compare.

Results (ρ = −0.30)

- **American Call (Heston): 3.3642**
- **European Call (Heston): 3.3966** (± 0.0282)

With no dividends, early exercise of a call is typically sub-optimal, so the American value should be at most only slightly above the European price and in practice they're nearly the same.

2.1.2 Delta and Gamma for the American Call Option

We calculated the Greeks for the American call we priced using Heston model, ($\rho = -0.30$): The result we got were $\Delta \approx 0.612$ and $\Gamma \approx 0.0720$.

In comparison to European call, when we calculated earlier, the Greeks were, $\Delta \approx 0.6178$ and $\Gamma \approx 0.0584$.

- Delta: the American Δ is slightly lower (~0.006), which is reasonable as early-exercise flexibility can make the value a little less sensitive to small moves in the underlying price.

- Gamma: the American Γ is higher (~0.014). Extra early-exercise optionality increases curvature (convexity) relative to the European, and LSM's exercise boundary can accentuate this around the money. Again, the difference is modest but directionally consistent with the American's added optionality.

Bottom line is, the Greeks we got for American options are very close to the European's (as theory suggests with no dividends), with a marginally lower Δ and higher Γ for the American due to early-exercise optionality and small LSM estimation effects.

2.2 Barrier Options

Barrier Options are a class of exotic, path-dependent options whose payoff (or even existence) depends on whether the underlying asset's price reaches a specified barrier level during the life of the option. If the barrier is breached, the option can either be activated ("knock-in") or terminated ("knock-out") before expiry (Rodrigo, 2020).

2.2.1 European CUI Using Heston

Using the Heston model data from Question 6 ($\rho=-0.70$), we priced a European up-and-in call with barrier H=95 and strike K=95. An up-and-in option is a knock-in barrier: it starts out "asleep" and only becomes active if, at any time before maturity, the stock touches or crosses the barrier. If the barrier is never hit, the option expires worthless even if the terminal price ends above the strike. If it is hit, the option behaves like a plain European call and pays max (S_T − K, 0) at expiry.

In code, we simulate many Heston price paths over time with $\rho=-0.70$. For each path we check whether the barrier was ever reached:

We then apply the call payoff at maturity only on 'hit' paths, zero otherwise, and discount–average to get the CUI price. For context, we also compute the plain European call with the same strike (no barrier), which is just the discounted average of max (S_T − K, 0) across all paths.

Because the barrier here sits above the current spot (H=K=95 vs S_0=80), not all paths will knock in, so the up-and-in call is cheaper than the plain call. But since barrier and strike are the same and close to the payoff region, many paths that finish in-the-money will have crossed 95 on the way, so the two prices are close exactly what we observe in the results.

The result makes sense and matches the economic intuition for barrier options:

The European up-and-in call with barrier and strike at 95 was priced at 0.0206, compared to 0.0221 for the plain European call. The knock-in condition makes it slightly cheaper, since the option only becomes active if the barrier is reached, but the difference is tiny (~0.0015). This is because most in-the-money paths already touch the barrier, so the two options behave almost identically. Overlapping confidence intervals confirm that the prices are statistically indistinguishable.

2.2.2 European PDI Using Heston

Using the Merton jump-diffusion setup from Question 8 (λ=0.75), we priced a European down-and-in put with barrier H=65 and strike K=65. A down-and-in option is also a knock-in barrier: it starts out inactive and only becomes alive if the underlying asset's price falls to or below the barrier at some point before maturity. If the barrier is never touched, the option expires worthless; if it is touched, the option behaves like a regular European put and pays max (K − S_T, 0) at expiry.

To capture this, we simulate full price paths under the Merton model, where prices evolve with both continuous diffusion and random jumps. For each path, we check if the stock ever dropped to or below 65. If so, we allow the usual put payoff at maturity; otherwise, the payoff is zero. Discounting and averaging across all paths gives the down-and-in put price. For comparison, we also calculate the plain European put with the same strike, which is always alive and so requires no barrier check.

Results:

- European down-and-in put (H=65, K=65, λ=0.75): 2.7426 (±0.0468)
- Plain European put (K=65, λ=0.75): 2.7569

As expected, the barrier feature makes the down-and-in put slightly cheaper than the plain put, since it knocks in only if the asset touches the barrier. However, because the strike and barrier coincide and are relatively close to the current spot, most scenarios where the put finishes in the money already involve the asset hitting the barrier along the way. That makes the two prices nearly identical; the small difference (~0.014, less than 1%) lies well within the simulation error band, confirming that in practice the barrier has little impact in this setup.

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